

Ideation Phase

Define the Problem Statements

Date	29 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	2 Marks

Customer Problem Statement Template:

"Plugging into the Future: An Exploration of Electricity Consumption Patterns" focuses on understanding the challenges faced by policymakers, energy providers, and consumers in analyzing electricity usage effectively. The project transforms complex electricity consumption data into clear, interactive visualizations that reveal trends, regional differences, and the effects of significant events such as the COVID-19 pandemic. By providing meaningful insights, it supports informed decision-making, promotes efficient energy management, and helps stakeholders develop sustainable strategies to meet future electricity demands.

I am	Describe customer with 3–4 key characteristics – <i>who are they?</i>	Describe the customer and their attributes here
I'm trying to	List their outcome or "job" the core about – <i>what are they trying to achieve?</i>	List the thing they are trying to achieve here
but	Describe what problems or barriers stand in the way – <i>what bothers them most?</i>	Describe the problems or barriers that get in the way here
because	Enter the "root cause" of why the problem or barrier exists – <i>what needs to be solved?</i>	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions from the customer's point of view – <i>how does it impact them emotionally?</i>	Describe the emotions the result from experiencing the problems or barriers

Example:

I am	I'm trying to	but	because	which makes me feel
a homeowner / tenant / energy manager	understand and manage my electricity consumption effectively to save cost and energy	it is difficult to track usage patterns, identify high consumption periods, and compare usage	data is complex, scattered across bills and devices, and lacks clear visual insights	confused, worried about high bills, and unsure how to reduce usage

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A government energy planner.	Monitor electricity consumption trends across different states and time periods.	I find it difficult to identify consumption patterns and sudden changes using raw data.	The electricity data is large, complex, and spread across multiple regions and months.	Concerned about making effective energy planning and policy decisions.
PS-2	A power distribution company manager.	Predict electricity demand and manage power supply efficiently.	I cannot easily identify seasonal demand variations and unexpected changes in consumptions.	There is no clear visual analysis to compare historical electricity usage patterns.	Uncertain about resource allocation and maintaining a stable power supply.